

MAM2000W Module 2LA: Linear Algebra

Chapter 2

Some answers to the exercises

2023

Section 2.1

1. (a) No (b) No (c) Yes (d) Yes (e) No

We provide proofs for (a) and (d) below.

(a) Axiom (8) does not hold: $1 \cdot (4, 5) = (8, 10)$, which is not the same as $(4, 5)$.

(d) Let $V = \mathbb{R}^+$ with addition defined by $x + y = xy$ for all $x, y \in \mathbb{R}^+$ and scalar multiplication defined by $\lambda x = x^\lambda$ for all $x \in \mathbb{R}^+$ and $\lambda \in \mathbb{R}$. [Note: In the perhaps rather confusing looking expression $x + y = xy$, the $+$ indicates the addition operation in this vector space; xy indicates usual product of the real numbers x and y .]

V is closed under addition, because, for any two positive real numbers x and y , xy is again a positive real number.

V is closed under scalar multiplication because, for any scalar (i.e. arbitrary real number) k and any positive real number x , x^k is again a positive real number.

The following hold for any $x, y, z \in \mathbb{R}^+$ and any $r, s \in \mathbb{R}$.

(1) $x + y = xy = yx = y + x$ (Here we use the fact that real number multiplication is commutative.)

(2) $(x + y) + z = (xy)z = x(yz) = x + (y + z)$

(3) $x + 1 = x \cdot 1 = x$, so 1 is a zero vector. [Note that $1 \in \mathbb{R}^+$.]

(4) $x + \frac{1}{x} = x \cdot \left(\frac{1}{x}\right) = 1$, so a negative of x is $\frac{1}{x}$. [Note that again we

are using $x \in \mathbb{R}^+$; then $\frac{1}{x} \in \mathbb{R}^+$ also.]

(5) $r(x + y) = (xy)^r = x^r y^r = rx + ry$

(6) $(r + s)(x) = x^{(r+s)} = x^r \cdot x^s = rx + sx$

(7) $(rs)(x) = x^{(rs)} = (x^s)^r = r(sx)$

(8) $1(x) = x^1 = x$.

2. When you are checking the axioms for Example 2.1.12, keep in mind that $f = g$ if and only if $f(x) = g(x)$ for every $x \in [a, b]$. You will find that mostly an axiom holds in this vector space because a corresponding axiom holds for real numbers (e.g. $f + g = g + f$ because $f(x) + g(x) = g(x) + f(x)$ for every $x \in [a, b]$, since addition of real numbers is commutative).

3. No. Not closed under addition or scalar multiplication.

4. The given set is closed under the given operations of addition and scalar multiplication. For the eight axioms: (1) Yes (2) Yes (3) Yes (4) Yes (5) No (6) No (7) Yes (8) Yes

5. If $\mathbf{v}$ and $\mathbf{w}$ are both negative vectors of $\mathbf{u}$, we have $\mathbf{u} + \mathbf{v} = \mathbf{0}$, $\mathbf{u} + \mathbf{w} = \mathbf{0}$.
Now show $\mathbf{v} = \mathbf{w}$; start with $\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{u} + \mathbf{v}) = \dots$.
6. To prove: Theorem 2.1.15(c). If λ is a scalar, and $\mathbf{0}$ the zero vector in a vector space:

$$\begin{aligned}
 \lambda(\mathbf{0} + \mathbf{0}) &= \lambda(\mathbf{0}) && \text{by axiom (3)} \\
 \text{so } \lambda\mathbf{0} + \lambda\mathbf{0} &= \lambda\mathbf{0} && \text{by axiom (5)} \\
 \text{so } (\lambda\mathbf{0} + \lambda\mathbf{0}) + (-\lambda\mathbf{0}) &= \lambda\mathbf{0} + (-\lambda\mathbf{0}) && \text{by axiom (4)} \\
 \text{so } \lambda\mathbf{0} + \mathbf{0} &= \mathbf{0} && \text{by axioms (2),(4)} \\
 \text{so } \lambda\mathbf{0} &= \mathbf{0} && \text{by axiom (3)}
 \end{aligned}$$

For (d), Use the hint and the fact that the negative of a vector is unique.

7. Show that $(a - b)\mathbf{u} = \mathbf{0}$, then use Theorem 2.1.15(e).

Section 2.2

1. (a) No: $(9, 7, 5, 3) \in S$ but $(-9, -7, -5, -3) \notin S$.
(b) Yes (c) Yes (d) No (e) Yes (f) No (g) Yes
2. (a) Not a subspace (b) A subspace.
3. (a) A subspace (b) Not necessarily a subspace (c) Not a subspace.
(See solution to Question 5 of Tutorial 7 for (a) and (b).)
4. $\mathbf{0} = 0\mathbf{v}$ for any $\mathbf{v}$ in the subset.

Section 2.3

1. Yes
2. (a) Yes (b) No (c) Yes (d) No (e) Yes
3. Yes
4. Yes
5. No.

Consider an arbitrary polynomial $ax^2 + bx + c$ of P_2 . We check whether there exist real numbers t_1, t_2, t_3 such that

$$t_1(x + 1) + t_2(x^2 + x) + t_3(x^2 + 2x + 1) = ax^2 + bx + c.$$

Compare coefficients of x^2, x and 1 on either side, and write out the corresponding system of three linear equations.

Its augmented matrix is $\begin{pmatrix} 0 & 1 & 1 & a \\ 1 & 1 & 2 & b \\ 1 & 0 & 1 & c \end{pmatrix}$.

After Gauss reduction we obtain $\begin{pmatrix} 1 & 1 & 2 & b \\ 0 & 1 & 1 & a \\ 0 & 0 & 0 & a + c - b \end{pmatrix}$.

For this system to have a solution, we must have $a + c - b = 0$. So only those polynomials $ax^2 + bx + c$ satisfying $a + c - b = 0$ will be in the span of the given set. For instance, $x^2 + x + 1$ is not in the span of the given set.

Section 2.4

- (a) Show that the equation

$$\lambda_1(1, 1, 0, 0) + \lambda_2(0, 0, 1, 1) + \lambda_3(3, 4, 5, 6) = (0, 0, 0, 0)$$

has only the solution $\lambda_1 = 0, \lambda_2 = 0, \lambda_3 = 0$.

- (b) Show that the equation

$$\lambda_1 3 + \lambda_2(3 + x) + \lambda_3(3 + x + x^2) = 0 + 0x + 0x^2$$

has only the solution $\lambda_1 = 0, \lambda_2 = 0, \lambda_3 = 0$,

- (a) False. For example, $\{(1, 0), (0, 1), (0, 4)\}$ is linearly dependent (why?) but $\{(1, 0), (0, 1)\}$ is independent (why?).

(b) True. Suppose that $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is independent and $T = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a subset of S , for some $1 \leq k \leq n$. We prove that T is independent. So, suppose there are scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ such that $\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_k \mathbf{u}_k = \mathbf{0}$. Then $\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_k \mathbf{u}_k + 0 \cdot \mathbf{u}_{k+1} + \dots + 0 \cdot \mathbf{u}_n = \mathbf{0}$ also. Since S is independent, all the scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ are equal to zero, as required.

- Four vectors in $\mathbb{R}^3$ will always be linearly independent. Show that the equation

$$(1, 2, 3) = \lambda_1(1, -3, 4) + \lambda_2(2, -2, -1) + \lambda_3(-1, -5, 14)$$

has no solutions.

- Yes. No.

- Suppose that, in any vector space, we have $\mathbf{w}_1 = \mathbf{v}_1$, $\mathbf{w}_2 = \mathbf{v}_1 + \mathbf{v}_2$ and $\mathbf{w}_3 = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3$. Two things must be proved: (i) that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ independent implies $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ independent, and (ii) that $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ independent implies $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ independent. We prove (ii), and leave (i) to you. So, assume that $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is

independent. Suppose there are scalars $\lambda_1, \lambda_2, \lambda_3$ such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}$. Express the $\mathbf{v}$'s in terms of the $\mathbf{w}$'s to get:

$$\lambda_1(\mathbf{w}_1) + \lambda_2(\mathbf{w}_2 - \mathbf{w}_1) + \lambda_3(\mathbf{w}_3 - \mathbf{w}_2) = \mathbf{0}.$$

Since $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is independent, the coefficients of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ must all be zero. So $\lambda_1 - \lambda_2 = 0$ and $\lambda_2 - \lambda_3 = 0$ and $\lambda_3 = 0$. Solving this simple linear system of three equations in three unknowns gives $\lambda_1 = \lambda_2 = \lambda_3 = 0$, as required.

6. Use Theorem 2.4.9(a) and the fact that vectors are linearly independent if and only if they are not linearly dependent.
7. (a) The vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ are *linearly dependent* iff
 $(\exists \lambda_1 \in \mathbb{R})(\exists \lambda_2 \in \mathbb{R}) \dots (\exists \lambda_n \in \mathbb{R})$
 $[(\lambda_1 \neq 0 \vee \lambda_2 \neq 0 \vee \dots \vee \lambda_n \neq 0) \wedge (\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n = \mathbf{0})]$.
 (b) The vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ are *linearly independent* iff
 $(\forall \lambda_1 \in \mathbb{R})(\forall \lambda_2 \in \mathbb{R}) \dots (\forall \lambda_n \in \mathbb{R})$
 $[\neg(\lambda_1 \neq 0 \vee \lambda_2 \neq 0 \vee \dots \vee \lambda_n \neq 0) \vee \neg \text{the comprehension question}(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n = \mathbf{0})]$
 iff
 $(\forall \lambda_1 \in \mathbb{R})(\forall \lambda_2 \in \mathbb{R}) \dots (\forall \lambda_n \in \mathbb{R})$
 $[(\lambda_1 = 0 \wedge \lambda_2 = 0 \wedge \dots \wedge \lambda_n = 0) \vee \neg(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n = \mathbf{0})]$
 iff
 $(\forall \lambda_1 \in \mathbb{R})(\forall \lambda_2 \in \mathbb{R}) \dots (\forall \lambda_n \in \mathbb{R})$
 $[(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n = \mathbf{0}) \implies (\lambda_1 = 0 \wedge \lambda_2 = 0 \wedge \dots \wedge \lambda_n = 0)]$.
 8. (a) See Definition 2.5.25(b)
 (b) The problem is that we cannot write down linear combinations of infinitely many vectors. The statement will be true if we replace “linear combination of the others” by “a linear combination of a finite number of the others”.
 (c) Show that for every n , the set $S_n\{p_k : k = 0, 1, \dots, n\}$ is linearly independent and then use Exercise 3(b).

Section 2.5

1. (a) Not linearly independent (3 vectors in $\mathbb{R}^2$).
 (b) Does not span $\mathbb{R}^3$.
 (c) Does not span P_2 .
2. (a) A basis (b) Not a basis.
3. (a) $(\frac{5}{28}, \frac{6}{28})$ (b) $(1, -2, 3)$
4. A basis for the solution space is $\{(4, 1, 0, 0), (-3, 0, 1, 0), (1, 0, 0, 1)\}$; its dimension is 3.

5. A basis for the required subspace is $\{(1, 1, 0), (0, 0, 1)\}$; its dimension is 2.
6. There are many possibilities; the vector $(1, 0, 0)$ is one.
7. 2
8. Dimensions are (a) 9 (b) 3 (c) 6.
10. See Question 5 in Tutorial 9.

Section 2.6

1. (b) Since the first p columns of A form a basis for $CS(A)$, we can express any column of A (in particular, the k 'th one) as a linear combination of the first p columns of A . So suppose

$$x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix} + \dots + x_p \begin{pmatrix} a_{1p} \\ \vdots \\ a_{np} \end{pmatrix} = \begin{pmatrix} a_{1k} \\ \vdots \\ a_{nk} \end{pmatrix},$$

for some $x_1, \dots, x_p$.

Consider the row operation that replaces Row i by Row $i + c$ Row ℓ . This only affects the i th row of A . The i th entry on the left-hand side becomes:

$$\begin{aligned} &= x_1(a_{i1} + ca_{\ell 1}) + \dots + x_p(a_{ip} + ca_{\ell p}) \\ &= (x_1a_{i1} + \dots + x_pa_{ip}) + c(x_1a_{\ell 1} + \dots + x_pa_{\ell p}) \\ &= a_{ik} + ca_{\ell k}. \end{aligned}$$

This is precisely what you get by applying the given row operation to the k th column of A .

Do the other types of row operation in a similar way.

2. (a) nullity(A)=1 and rank(A)=2
 (b) nullity(B)=2 and rank(B)=2
 (c) nullity(C)=2 and rank(C)=3
3. (a) 4 and 0 (b) 3 and 2 (c) 3 and 0
4. (a) No; no (b) No; yes (Use Theorem 2.6.20.)
5. Use theorem 2.6.20 and the fact that a square matrix is invertible if and only if it is non-singular.

6. (a) First note that $NS(B)$ and $NS(AB)$ are both subsets of $\mathbb{R}^p$. Next, we check that $NS(B)$ is a subset of $NS(AB)$: if $\mathbf{x} \in NS(B)$ then $B\mathbf{x} = \mathbf{0}$, so $A(B\mathbf{x}) = \mathbf{0}$, so $(AB)\mathbf{x} = \mathbf{0}$; so we have $\mathbf{x} \in NS(AB)$. The fact that $NS(B)$ is closed under addition and scalar multiplication is trivial, because we have already checked that null spaces of matrices are always subspaces of $\mathbb{R}^p$ (and the addition and scalar multiplication operations here are the usual ones).
- (b) From (a) above, $\text{nullity}(B) \leq \text{nullity}(AB)$. We now apply the dimension theorem for matrices, and use the fact that B and AB both have p columns to get $p - \text{rank}(B) \leq p - \text{rank}(AB)$, and so $\text{rank}(AB) \leq \text{rank}(B)$ as required.
7. The nullities are 1, 2, $n - 2$, respectively (for $n \geq 2$). The ranks are 2, 2, 2, respectively (again for $n \geq 2$).
8. (a) and (b) 5 vectors in $\mathbb{R}^3$ in both cases.
- Note:** For questions 9(a) and 10(a), the intention was that not *all* the vectors should be row or column vectors of the given matrix. Also, there are clearly many correct answers for questions 9 and 10.
9. (a) $\{(1, 0, -1), (0, 1, 0)\}$
 (b) $\{(1, 2, -1), (1, 9, -1)\}$

$$10. \text{ (a) } \left\{ \begin{pmatrix} 1 \\ 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 8 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 5 \\ 3 \end{pmatrix} \right\}$$

$$\text{ (b) } \left\{ \begin{pmatrix} 1 \\ 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 5 \\ 3 \end{pmatrix} \right\}$$

Revision exercises

1. (a) Yes (b) No.
2. (a) Yes (b) No (c) Yes.
3. (a) $U + W \neq \emptyset$, since $\mathbf{0} \in U, \mathbf{0} \in W$ and therefore $\mathbf{0} = \mathbf{0} + \mathbf{0} \in U + W$. Let $\mathbf{v}_1, \mathbf{v}_2 \in U + W$. Then there are $\mathbf{u}_1, \mathbf{u}_2 \in U, \mathbf{w}_1, \mathbf{w}_2 \in W$ such that $\mathbf{v}_1 = \mathbf{u}_1 + \mathbf{w}_1, \mathbf{v}_2 = \mathbf{u}_2 + \mathbf{w}_2$. Hence $\mathbf{v}_1 + \mathbf{v}_2 = (\mathbf{u}_1 + \mathbf{w}_1) + (\mathbf{u}_2 + \mathbf{w}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{w}_1 + \mathbf{w}_2) \in U + W$, since U and W are subspaces and hence closed under addition. This shows that $U + W$ is closed under addition.
- Let λ be a scalar and $\mathbf{v}_1 \in U + W$ (as above). Then $\lambda\mathbf{v}_1 = \lambda(\mathbf{u}_1 + \mathbf{w}_1) = \lambda\mathbf{u}_1 + \lambda\mathbf{w}_1 \in U + W$, since U and W are closed

under scalar multiplication. Hence $U + W$ is closed under scalar multiplication.

It follows that $U + W$ is a subspace.

- (b) If $\mathbf{v} \in U \cup W$, then $\mathbf{v} \in U$ or $\mathbf{v} \in W$. Suppose $\mathbf{v} \in U$, then $\mathbf{v} = \mathbf{v} + \mathbf{0} \in U + W$. It follows that $U \cup W \subseteq U + W$, and so, by (a), $U + W$ is a subspace containing $U \cup W$. If S is another such subspace and $\mathbf{v} \in U + W$, then there are $\mathbf{u} \in U, \mathbf{w} \in W$ such that $\mathbf{v} = \mathbf{u} + \mathbf{w}$. Then, since $U \cup W \subseteq S$, $\mathbf{u}, \mathbf{w} \in S$, and since S is a subspace, $\mathbf{v} = \mathbf{u} + \mathbf{w} \in S$. This shows that $U + W \subseteq S$. (It follows that $U + W$ is contained in any subspace S containing $U \cup W$, and is therefore the smallest such subspace.)
- (c) Let $\mathbf{v} \in V = U + W$. Then there are $\mathbf{u} \in U, \mathbf{w} \in W$ such that $\mathbf{v} = \mathbf{u} + \mathbf{w}$. Suppose also that $\mathbf{v} = \mathbf{u}_1 + \mathbf{w}_1$, with $\mathbf{u}_1 \in U, \mathbf{w}_1 \in W$. Then $\mathbf{u} + \mathbf{w} = \mathbf{u}_1 + \mathbf{w}_1$, and so $\mathbf{u} - \mathbf{u}_1 = \mathbf{w}_1 - \mathbf{w}$. But $\mathbf{u} - \mathbf{u}_1 \in U, \mathbf{w}_1 - \mathbf{w} \in W$. Since $U \cap W = \{\mathbf{0}\}$, we must have $\mathbf{u} - \mathbf{u}_1 = \mathbf{0} = \mathbf{w}_1 - \mathbf{w}$, and so $\mathbf{u} = \mathbf{u}_1, \mathbf{w} = \mathbf{w}_1$.
- (d) Let W be the set of all constant functions on $[0, 1]$. Then W is a subspace of V (it is non-empty and clearly closed under addition and scalar multiplication). If $f \in V$, let g and h be the continuous functions defined by $g(x) = f(x) - f(1)$ and $h(x) = f(1)$ for all $x \in [0, 1]$. Then $g(1) = f(1) - f(1) = 0$, so that $g \in U$, and clearly $h \in W$. But $f(x) = f(x) - f(1) + f(1) = g(x) + h(x)$ for all $x \in [0, 1]$, so $V = U + W$. If $f \in U \cap W$, then f is a constant function and $f(1) = 0$, so $f(x) = 0$ for all $x \in [0, 1]$. Hence $U \cap W = \{\mathbf{0}\}$ and so $V = U \oplus W$.

4. (a) True (b) False (c) True.